Spring

The Boundaries of Spring:
Mathematical Limits of Auditability and Non-Auditability

SCX

202671

v1.0 — — SCX — Spring

Abstract

SPRING—— SPRING (1) **Hoeffding**——Spring (2) **Gödel**——Spring (3) ≠Korzybski——Spring **Boundary Theorem** SpringSpring$1 - e^{-2M\Delta^2}$ ————Spring Spring
Spring—— Spring GPTSpring
SpringHoeffdingGödel- SCX

Contents

1

1.1

””capability

SPRING——SPRING

□

1.2 Spring

SPRING [?]”” Training-Is-Audit GPTClaudeGemini$M = 1$UNDECLARED SPRING——M_t CERCISSPRING**born audited**——

——Hoeffding——

SPRING

1.3

1 **Hoeffding** $\mathbb{P}(\text{error}) \leq e^{-2M\Delta^2}$ $M \to \infty$00 Spring——Spring

2 **Gödel** Spring Gödel Spring——M

3 $\neq$ DFTYAJIE $=$ max13 Spring

Hoeffding, 1963 [?] Gödel, 1931 [?] Korzybski, 1933 [?] Spring—— SpringSpring

1.4

Boundary Theorem

Definition 1.1 (Spring). $\mathcal{F}_{\text{SPRING}}$*Spring*——*Spring*

- Auditable(SPRING) $= \{\varphi \in \mathcal{F}_{\text{SPRING}} \mid \text{SPRING } \varphi\ \}$
- (SPRING) $= \mathcal{F}^c_{\text{SPRING}} \cup \{\varphi \in \mathcal{F}_{\text{SPRING}} \mid \varphi\ \}$

∂_{Spring} SPRING——

——HoeffdingGödelKorzybski

1.5

(1) 21——Hoeffding

(2) 32——Gödel

(3) 43——$\neq$

(4) 5——

(5) 6Spring——

(6) 7Spring——

(7) 8——GPT/Claude/Gemini

(8) 9——

2 1Hoeffding

2.1

SpringHoeffding [?] SCX1 [?]

$$\mathbb{P}(\text{error}) \leq e^{-2M\Delta^2}, \tag{1}$$

$M\Delta$

M Spring—— MM

$M0$

2.2

Theorem 2.1 (Hoeffding). *$M \in \mathbb{N}$SCX1 (A1–A6)* Spring

$$R(M,\Delta) = \mathbb{P}(\mathit{error} \mid M,\Delta) > 0. \tag{2}$$

$M < \infty$

$$R(M,\Delta) \geq \inf \mathbb{P}(\mathit{error}) > 0. \tag{3}$$

Spring*Hoeffding*

$$0 < R(M,\Delta) \leq e^{-2M\Delta^2}. \tag{4}$$

. []$R(M,\Delta) \leq e^{-2M\Delta^2}$Hoeffding MBernoulli——SCX1 [?]

$R(M,\Delta) > 0$ 0

SCX1(A6)$\eta > 0$ ——

Ms p_0p_1

$$\omega^* = \{(x,y)\ \forall i:\ i\ (x,y)\ \} \tag{5}$$

$\mathbb{P}(\omega^*) \geq \eta \cdot \min(p_0, 1-p_1)^M > 0$

———— Q_0Q_1 SCX3

$\min(\mathbb{P}(\text{type I error}), \mathbb{P}(\text{type II error})) > 0$

$\omega^* R(M,\Delta) \geq \mathbb{P}(\omega^*) \cdot () > 0$

□

Remark 2.2 (”$\lim R = 0$”). $M \to \infty$Hoeffding$e^{-2M\Delta^2} \to 0$ 0——Spring$M$ ”””” $\lim_{n\to\infty}(1-1/n)^n = e^{-1} \neq 1$

2.3

Hoeffding

Corollary 2.3 (Spring)**.** SPRING SPRING*confidence intervalabsolute guarantee* ——*Hoeffding*——

$$\boxed{\text{Spring}X\ 1 - e^{-2M\Delta^2}\ X}$$

GPT"500K"—— SPRING $+ + M + \Delta +$ """""——
SpringSpringGPT"" Spring—— Hoeffding

2.4

100%

(1) $M = 20\Delta = 0.3\ e^{-2\cdot 20\cdot 0.09} = e^{-3.6} \approx 0.027$——97.3% ""

(2) 97.3%SPRING97.3% Hoeffding $M\Delta$

(3) ——SPRING100% bug SPRING

3 2Gödel

3.1

1—— SPRING SPRINGYAJIE ””””
SPRING SPRINGGödel

3.2

SPRING

Definition 3.1 (Spring). $\mathcal{F}_{\textsc{Spring}}$SPRING

- DFT
- $\wedge, \vee, \neg, \implies$
- $\forall\exists$
- $E_1 < E_2\|\mathbf{f}_1\| < \|\mathbf{f}_2\|$

SPRING

$$\mathcal{A}_{\textsc{Spring}} : \mathcal{F}_{\textsc{Spring}} \to \{0, 1, \mathit{UNDECIDED}\}, \tag{6}$$

0”” 1”” $\geq 1 - e^{-2M\Delta^2}$ UNDECIDED””

Remark 3.2 ($\mathcal{F}_{\textsc{Spring}}$). $\mathcal{F}_{\textsc{Spring}}$ $\mathrm{F}_{\textsc{Spring}}||\forall E < E'$
$\mathcal{F}_{\textsc{Spring}}$Gödel””

3.3

Theorem 3.3 (SpringGödel). $\mathcal{F}_{\textsc{Spring}}\varphi_{\textsc{Spring}} \in \mathcal{F}_{\textsc{Spring}}$

(i) $\varphi_{\textsc{Spring}}\varphi_{\textsc{Spring}}$

(ii) $\mathcal{A}_{\textsc{Spring}}(\varphi_{\textsc{Spring}}) = \mathit{UNDECIDED}$—— SPRING$\varphi_{\textsc{Spring}}M$

$\varphi_{\textsc{Spring}}$——SPRING*Gödel*

. []Gödel [**?**]$\mathcal{F}_{\textsc{Spring}}$
1 $\mathcal{F}_{\textsc{Spring}}$ Gödel DFTZ**r** $\mathbf{a}_i$ YAJIE
2 $\mathcal{F}_{\textsc{Spring}}$

$$\mathrm{Provable}_{\textsc{Spring}}(x) \iff x\ \mathcal{F}_{\textsc{Spring}}\ \textsc{Yajie} \tag{7}$$

diagonal lemma [**?**]

3Gödel

$$\varphi_{\textsc{Spring}} \equiv \text{“}\varphi_{\textsc{Spring}}\ \mathcal{F}_{\textsc{Spring}}\ \text{”} \tag{8}$$

$\varphi_{\textsc{Spring}} \equiv \neg\mathrm{Provable}_{\textsc{Spring}}(\ulcorner\varphi_{\textsc{Spring}}\urcorner)$

4

- $\mathcal{A}_{\textsc{Spring}}(\varphi_{\textsc{Spring}}) = 1$ $\mathrm{Provable}_{\textsc{Spring}}(\ulcorner\varphi_{\textsc{Spring}}\urcorner)$ $\neg\varphi_{\textsc{Spring}}$——$\varphi_{\textsc{Spring}}$
- $\mathcal{A}_{\textsc{Spring}}(\varphi_{\textsc{Spring}}) \neq 1$ $\neg\mathrm{Provable}_{\textsc{Spring}}(\ulcorner\varphi_{\textsc{Spring}}\urcorner)$——$\varphi_{\textsc{Spring}}$ $\mathcal{A}_{\textsc{Spring}}(\varphi_{\textsc{Spring}})$10—— UNDECIDED

5 Gödel

”$V_\theta \mathbf{x}$ $|V_\theta(\mathbf{x}) - E_{\mathrm{DFT}}(\mathbf{x})| < \varepsilon$ V_θSPRING$M = 100$” SPRING Gödel$\mathcal{F}_{\textsc{Spring}}$ □

3.4

Corollary 3.4. ——SPRING——

SPRING ***Gödel***

SPRING

- **DFT** Kohn-Sham-
-
-
- ε ε3

$\mathbf{g}_{\textsc{Spring}} = \mathbf{0}$ SPRING””SCX3 UNDECIDED

GödelSpring—— SpringGPTGPT

4 3$\neq$

4.1 Korzybski

Alfred Korzybski1933*Science and Sanity* [?]

$$\boxed{\text{The map is not the territory.}} \; \square$$

SPRINGSPRING

4.2 Spring

SPRING

1 DFT-PBE [?]SCAN [?]- ——

2

3 SPRING$V_\theta(\mathbf{x})$Born-Oppenheimer ——SPRING

4 k

$\varepsilon_{\text{phys}}$ **Spring**

4.3 M""

$y^* \hat{y} = y^* + \varepsilon_{\text{phys}}$
$M V_\theta^{(i)}$ DFT prediction$\hat{y} y^*$

$$\mathbb{E}[V_\theta^{(i)}(\mathbf{x})] = \hat{y}(\mathbf{x}) = y^*(\mathbf{x}) + \varepsilon_{\text{phys}}(\mathbf{x}), \tag{9}$$

$\varepsilon_{\text{phys}}(\mathbf{x})$——$M$

4.4

3

Theorem 4.1 ($\neq$). *$y^* \hat{y} V_{\text{consensus}}$*YAJIE SPRING

$$\varepsilon = \max\Big(\underbrace{\varepsilon_{stat}(M, \Delta)}_{1},\ \underbrace{\varepsilon_{phys}}_{3}\Big), \tag{10}$$

- *$\varepsilon_{stat}(M, \Delta)$Hoeffding1$M \to \infty \varepsilon_{stat} \to 0$*
- *ε_{phys}3$\varepsilon_{phys}M$ DFT*

$\varepsilon_{phys} > 0$

$$\lim_{M\to\infty} \varepsilon = \varepsilon_{phys} > 0. \tag{11}$$

. []**1** $1\varepsilon_{\text{stat}} \leq e^{-2M\Delta^2} \to 0$

2 $\varepsilon_{\text{phys}} = \sup_{\mathbf{x}} |\hat{y}(\mathbf{x}) - y^*(\mathbf{x})|$ DFT

——

3 YAJIE MYAJIE—— $\varepsilon_{\text{phys}}$

4 $\varepsilon = \max(\varepsilon_{\text{stat}}, \varepsilon_{\text{phys}})$ $\varepsilon_{\text{stat}} \to 0\varepsilon_{\text{phys}}$ □

4.5 Corollary: Spring

Corollary 4.2. SPRING

- SPRING*DFT PBE""*
- SPRING*""*
- SPRING*""""*

——SPRING

SpringDFT—— "AlNa=3.11Å"——Spring12=0.004Å 98.7%SpringDFT PBEAlN Springground truthDFT PBE """"""""

5

5.1

123

Theorem 5.1 (Spring —). $\mathcal{F}_{\text{SPRING}}$SPRING$\mathcal{A}_{\text{SPRING}}$SPRING SPRINGAuditable(SPRING)(SPRING) ∂_{Spring}

1 ***Hoeffding*** $\varphi \in \mathcal{F}_{\text{SPRING}}$ SPRING$1 - e^{-2M\Delta^2}$ $M\Delta$- *1MM*

$$\forall M < \infty : \mathbb{P}(\mathcal{A}_{\text{SPRING}}(\varphi)\ ) > 0. \tag{12}$$

2 ***Gödel*** $\varphi^{\perp}_{\text{SPRING}} \in \mathcal{F}_{\text{SPRING}}$ $\mathcal{A}_{\text{SPRING}}(\varphi^{\perp}_{\text{SPRING}}) = \textit{UNDECIDED}$ SPRING—— SPRING

$$\exists \varphi \in \mathcal{F}_{\text{SPRING}} : \ (\varphi\ ) \wedge (\mathcal{A}_{\text{SPRING}}(\varphi) = \textit{UNDECIDED}). \tag{13}$$

3 ***Korzybski*** SPRING$\varepsilon \geq \varepsilon_{phys}$ ε_{phys} SPRING

$$\lim_{M \to \infty} \varepsilon = \varepsilon_{phys} \not\to 0\ \varepsilon_{phys} = 0. \tag{14}$$

∂_{SPRING}SPRING

- SPRING$1 - e^{-2M\Delta^2}$
- *Gödel*SPRINGUNDECIDED——

5.2

Self-Awareness of the Boundary

Definition 5.2. SPRINGφSPRING

(1) $\varphi\mathcal{F}_{\text{SPRING}}$

(2) $\varphi\Delta M$

(3) φ

(4)

Proposition 5.3. SPRING

- $\mathcal{F}_{\text{SPRING}}$——
- *HoeffdingM*Δ——ΔM SPRING
- SPRING*"DFT""""* ——SPRING

Spring""GPT""—— ""/""Spring"76%"—— 76%100%

5.3

Spring

Protocol 5.4. Spring——ΔM——

(1) $\theta_{\min}$95% UNDECIDED

(2) "Spring [M///]"

(3) DFTCCSD(T)

(4) ——Spring

6 Spring

6.1

Spring Spring——

Spring

(1) $\mathcal{F}_{\textbf{Spring}}$ Spring—— DFT

(2) Spring/ $\Delta > 0$

(3) Spring ——DFT PBEDFT PBE

6.2

Spring

1 Spring$M$ $M\delta$ HoeffdingRMSE

x

$$\mathbb{P}\big(|V_{\text{consensus}}(\mathbf{x}) - \mathbb{E}[V(\mathbf{x})]| > \delta\big) \leq e^{-2M\Delta^2}. \tag{15}$$

——

2 SpringYajie SCX1M

ηF1 $\geq 1 - \frac{1}{\eta}\sum_s \rho_s e^{-2M\Delta_s^2}$

3 Spring "(111)"——DFT(111)

4 Spring""Query-Is-Computation ""—— SpringLLMSpring M_t

——

5 SpringCercis CercisYajieM_t

""Hoeffding

6 - Spring$\mathcal{H}(\mathcal{D}) = \mathcal{H}(\mathcal{D})$

$\mathcal{H}(\mathcal{D})$

6.3

Table 1: Spring

/	Hoeffding	1
	F1	1+3
		2
		—
		1
-		—

”” ””_____

7 Spring

7.1

SpringSpring—— Spring

7.2

1 SpringDFT"" SpringDFT

3——$\neq$ SpringDFT ""——Spring
UNDECIDED +

2 SpringACE [?]MACE [?] NequIP [?]""

2Gödel+ 3 ——
UNDECIDED +

3 **Gödel** Spring—— "SpringM"

2——Gödel SpringSpring
UNDECIDED + Gödel

4 Spring100%

1——Hoeffding M0 ——Spring
$1 - e^{-2M\Delta^2} < 100\%$ + 100%

5 Spring$\mathcal{F}_{\text{Spring}}$——

Spring Spring
OUT OF SCOPE——"Spring"

6 Spring—— $\mathcal{F}_{\text{Spring}}$

Δ- Hoeffding
INSUFFICIENT DATA + ΔM

7 Spring——

13 ——
SPECULATIVE +

8 Spring——

2 + 3——DFT Spring""""
+ "[]"

7.3

Table 2: Spring

	3	UNDECIDED	/
	2+3	UNDECIDED	/
Gödel	2	UNDECIDED	
	1	¡100%	——
		OUT OF SCOPE	
	1	INSUFFICIENT DATA	
	1+3	SPECULATIVE	
	2+3		

Table 3: *

......

::

8 GPTClaudeGemini

8.1 vs.

SpringGPT [?]Claude [?]Gemini [?] ——
""________

(1) $M = 1$ SCX2 GPT""——GPTGPT

(2) **UNDECLARED** —— $M\Delta$

(3) GPTClaudeGemini—— "PythonCSV" "" """"

(4)

____""

8.2

??

Table 4: Spring vs. —

	Spring	**GPT/Claude/Gemini**	
	$M \geq 8$	$M = 1$	vs.
	AUDITED	UNDECLARED	vs.
	Hoeffding	""	vs.
			vs.
	$\mathcal{F}_{\text{Spring}}$		vs.
Gödel			vs.
	UNDECIDED		vs.
/			vs.
	+		vs.
			vs.

8.3

"AlN500K"

GPT/Claude/Gemini

"AlN500Ka = 3.114 Åc = 4.986 Å "

-
- """"
-
-

Spring

AlN500Ka = 3.115 ± 0.003 Å, c = 4.991 ± 0.005 Å

- Yajie $M = 12$
- 97.3% (Hoeffding$e^{-2\cdot 12\cdot 0.15^2} \approx 0.027$)
- Cercis0.91/1.0
- DFT PBE + SpringACE
- -$\mathcal{H}(\mathcal{D})$ = a3f2...e71b
- 12- → → 500K
- DFT PBE$\varepsilon_{\text{phys}}$PBE vs.

- +
- 12
- 97.3%Hoeffding
- DFT PBE
- $\varepsilon_{\text{phys}}$

——**GPT Spring**

8.4

(1) GPTtoken——

(2) ——

(3) GPT"""" ——GPT """"—— ""

(4) **""** GPT""""——

""hallucination

——

9

9.1

Spring

1 Hoeffding Spring

2 Gödel SpringSpring Spring

3 $\neq$ Spring

Spring

Spring $1 - e^{-2M\Delta^2}$ Spring

9.2

$\square$

——Hoeffding

Spring

SpringBoundary Self-Awareness—— ———— ————

9.3 GödelSpring

Kurt Gödel1931

Hilbert

Gödel—— ”” ””

SpringGödel——HoeffdingKorzybski——

Spring—— Spring

$\boxed{\text{Gödel: }}$

$\boxed{\text{Spring: ——}}$

9.4

SpringAI

(1)

”””” ——GPT””””

(2)

—— Spring

(3) ——SpringAI—— ”” ””limitations section ——7

(4) —— —— ”” ””

9.5 SpringSpring

Alan Turing—— - ——

SpringSpring—— Spring Spring

$\boxed{\text{SpringSpring}}$

□

—— SCX, 202671

Hoeffding (1963) [?] Gödel (1931) [?]Korzybski (1933) [?] ""——""

Spring

SCX""Theorem 3""Theorem 2——

SCX

SCX

SCX[repository] Spring5

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A A

A.1 Spring

Definition A.1 (Spring —). $\textsc{Spring}\mathcal{A}_{\textsc{Spring}} = (\mathcal{F}_{\textsc{Spring}}, \mathcal{E}, \mathcal{P}, \mathcal{C}, \mathcal{B})$

- $\mathcal{F}_{\textsc{Spring}}$
- $\mathcal{E} = \{E_1, \ldots, E_M\} M$
- $\mathcal{P}$Yajie
- $\mathcal{C}$CercisM_t
- $\mathcal{B}$

A.2 Hoeffding

Definition A.2 (Hoeffding). $\varphi \in \mathcal{F}_{\textsc{Spring}}$ $C_i(\varphi) \in [0,1] E_i \varphi$

$$\bar{C}(\varphi) = \frac{1}{M} \sum_{i=1}^{M} C_i(\varphi). \tag{16}$$

$\varepsilon > 0$

$$\mathbb{P}(|\bar{C}(\varphi) - \mathbb{E}[\bar{C}(\varphi)]| \geq \varepsilon) \leq 2\exp(-2M\varepsilon^2). \tag{17}$$

A.3 Gödel

$\mathcal{F}_{\textsc{Spring}}$Gödel——

Example A.3 (SpringGödel). (1) $\mathrm{AuditPass}_M(x, \varphi)$ "$xM \geq 1 - e^{-2M\Delta^2} \varphi$"

(2) $d(\varphi) = \varphi(\ulcorner \varphi \urcorner)$

(3) $\varphi_{\textsc{Spring}} \equiv \neg \mathrm{AuditPass}_M(\ulcorner \varphi_{\textsc{Spring}} \urcorner, \varphi_{\textsc{Spring}})$
$\varphi_{\textsc{Spring}}$"Spring" $\varphi_{\textsc{Spring}}\mathrm{AuditPass}_M(\ulcorner \varphi_{\textsc{Spring}} \urcorner, \varphi_{\textsc{Spring}})$ $\varphi_{\textsc{Spring}}$ $\mathcal{A}_{\textsc{Spring}}(\varphi_{\textsc{Spring}}) = \text{UNDECIDED}$

A.4 -

Definition A.4 ($\varepsilon_{\text{phys}}$). y

- $y^* y$——
- $\hat{y}_{DFT} y DFT$
- $\varepsilon_{phys} = \sup |\hat{y}_{DFT} - y^*|$

Spring$\hat{y}_{DFT}$ $\mathbb{E}[V_\theta^{(i)}] = \hat{y}_{DFT} \neq y^* \varepsilon_{phys} > 0$

B B

	English
Spring	Spring is the most auditable framework ever built.
Spring	Precisely because it is rigorous, Spring can identify its own fundamental limits.
——MSpring	The audit error is strictly positive — for any finite M, knowledge under Spring is always probabilistic.
Spring	Spring can give confidence intervals, not guarantees.
——	This is not a flaw — it is the mathematical nature of empirical knowledge.
	Any sufficiently strong formal system contains true but unprovable propositions.
Spring——	Spring must declare its axiomatic boundaries — which axioms it assumes, which it cannot prove.
	The map is not the territory.
$M\varepsilon$Yajie——ε	If all M experts agree within approximation error ε, Yajie returns HIGH confidence — but the territory may differ from the map by up to ε.
= max(1, 3)	The audit error = max(statistical error from Ceiling 1, systematic error from physical approximation).
SpringSpring$1 - e^{-2M\Delta^2}$	For claims within Spring's formal system, Spring provides an audit with confidence $1 - e^{-2M\Delta^2}$.

	English
Spring	For claims outside the formal system, Spring is silent.
Spring	The boundary is self-aware: Spring knows what it can audit, what it cannot, and why.
GPTSpring	GPT doesn't know its own limits. Spring does. That is the difference between a model and an audit framework.
	The strongest framework is not the one that claims it can audit everything — it is the one that knows precisely what it cannot audit.
Gödel:	Gödel: This is true, but I cannot prove it.
Spring: ——	Spring: This may be true, but I cannot audit it — here is why.
	Silence is a capability, not a flaw.
	A system operating without self-awareness is blind. A system operating with boundary self-awareness is awake.
Spring	Spring audits what it can audit, and precisely declares what it cannot. That is the entire difference between an audit framework and a hallucination generator.

Table 5: Spring